

Elias Ramzi — AI Research Scientist

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End-to-end driving — planning, world models, and multimodal models

Experiences

valeo.ai (Paris, France) - full-time

2024-

AI research scientist

Deep learning research interests: end-to-end planning, world models, VLMs / LLMs.

- Developed world action models for autonomous driving, VaViM/VaVAM (technical report, CoRL Workshop, initial baseline for AlpaSim).
- Scaled training of generative models up to 1.9B parameters across 192 H100s using μ P and DeepSpeed ZeRO-2.
- Built Pictura, a GPU-accelerated multi-agent driving simulator rendering every agent's egocentric view, which unlocks perspective-view self-play at scale for driving policies (ECCV 2026 workshop paper).
- Co-supervising two PhD students: one on LLMs and VLMs (ICLR 2025, ACL Findings 2026), another on world models and RL.

Cnam & Coexya (Paris, France) - 3 year PhD

2021-2024

PhD student

PhD focused on computer vision and image retrieval:

- Designed novel ranking losses for image retrieval, combining theoretical analysis with large-scale experiments.
- Published four papers in major international machine learning conferences (NeurIPS, ECCV $\times$ 2, ICML), three of which as first author, and 2 papers in journals (TPAMI, TMLR).
- Transferred HAPPIER (ECCV) to trademark logo retrieval; the resulting models were deployed in production in Coexya's Accepto product in 2024.
- Ran distributed training experiments across up to 16 GPUs on France's Jean Zay HPC cluster.

Cnam (Paris, France) - 5 month internship

2020

Research intern

Deep learning for 3D medical image segmentation. I used a learned confidence measure (ConfidNet) to combine segmentation results from different 2D U-Nets.

Education

PhD in Computer Science, Cnam & Coexya (France)

2021-2024

Ranking losses and hierarchical learning for image retrieval. Awarded the "Prix de Thèse" by AFRIF ☑ (June 2025). 6 publications including 4 first-author papers.

Supervisor: Nicolas Thome ☑ (Sorbonne Université).

Advisors: Nicolas Audebert ☑ (IGN); Clément Rambour ☑ (Sorbonne Université).

Industrial advisor: Xavier Bitot ☑ (Coexya).

Engineering diploma, CentraleSupélec (France)

2016-2020

Courses in signal processing, machine learning, statistical models, data science.

Technical skills

Open source projects

- Pictura simulator (ECCVw'26): 🌐 Pictura
- Scaling laws for driving video diffusion models (ECCVw'26): 🌐 VATIX
- Overleaf MCP server: 🌐 WebLatexMCP
- Reproduce VaViM and VaVAM (CoRL'25): 🌐 VideoActionModel
- Global and local prompts for VLMs (ECCV'24): 🌐 GalLoP
- Out-of-distribution detection with HEAT (ICML'23): 🌐 HEAT.
- Ranking losses and hierarchical image retrieval (NeurIPS'21, ECCV'22, TPAMI'25): 🌐 ROADMAP, 🌐 HAPPIER, 🌐 SupRank & the hierarchical 🌐 google-landmark dataset.

Technologies

- My programming language: Python – PyTorch, Lightning, NumPy, Hydra, Pandas, Scikit-learn.
- My work environment : Linux/macOS + Git + VS Code & claude-code.
- Other tools: L^AT_EX, Shell script, SLURM, HyperQueue, Nextflow.

Contributions to open-source

- PyTorch metric learning:  pytorch-metric-learning.
- Implementation of Mixture of Experts:  mixture-of-experts.
- PyTorch-Lightning:  lightning.
- Universal Image Embeddings:  Universal-Image-Embeddings.

Miscellaneous

- French is my mother tongue and I speak fluently English (last tested at C1 level).
- I enjoy biking ( ClaudeCyclingMCP), (trail) running and back-country skiing.

Publications

Yuan Yin, **Elias Ramzi**, Marc Lafon, Valentin Charraut, Victor Bares, Yihong Xu, Éloi Zablocki, Alexandre Boulch, Thibault Buhet, Andrei Bursuc, *et al.* "Pictura: Perspective-View Self-Play at Scale for Driving." *ECCVw*, 2026.

Victor Besnier, Anh-Quan Cao, **Elias Ramzi**, Spyros Gidaris, Tuan-Hung Vu, Andrei Bursuc, Éloi Zablocki, Matthieu Cord. "How Far Can 5,500 Hours of Driving Take You? A Scaling Law Analysis of Video Diffusion Models." *ECCVw*, 2026.

Yihong Xu, Éloi Zablocki, Yuan Yin, **Elias Ramzi**, Ellington Kirby, Alexandre Boulch, Matthieu Cord. "TOAD: Test-Time Trajectory Optimization for Autonomous Driving." *arXiv preprint*, 2026.

Shashanka Venkataramanan, Valentin Pariza, Mohammadreza Salehi, Lukas Knobel, Spyros Gidaris, **Elias Ramzi**, Andrei Bursuc, Yuki M. Asano, "Franca: Nested Matryoshka Clustering for Scalable Visual Representation Learning." *CVPR*, 2026.

Amaia Cardiel, Éloi Zablocki, **Elias Ramzi**, and Eric Gaussier. "DRIV-EX: Counterfactual Explanations for Driving LLMs." *ACL findings*, 2026.

Florent Bartoccioni, **Elias Ramzi**, *et al.* "VaViM and VaVAM: Autonomous Driving through Video Generative Modeling." *Technical report & CoRL workshop*, 2025.

Amaia Cardiel, Éloi Zablocki, **Elias Ramzi**, Oriane Siméoni, and Matthieu Cord. "LLM-wrapper: Black-Box Semantic Aware Adaptation of Vision-Language Models for Referring Expression Comprehension." *ICLR*, 2025.

Elias Ramzi, Nicolas Audebert, Clément Rambour, André Araujo, Xavier Bitot and Nicolas Thome. "Optimization of Rank Losses for Image Retrieval." *TPAMI*, 2025.

Marc Lafon, **Elias Ramzi**, Clément Rambour, Nicolas Audebert, Nicolas Thome. "GalLoP: Learning Global and Local Prompts for Vision-Language Models." *ECCV*, 2024.

Yannis Karmim, **Elias Ramzi**, Raphaël Fournier S'niehotta, and Nicolas Thome. "ITEM: Improving Training and Evaluation of Message-Passing based GNNs for top-k recommendation." *TMLR*, 2024.

Marc Lafon, **Elias Ramzi**, Clément Rambour, Nicolas Thome. "Hybrid Energy Based Model in the Feature Space for Out-of-Distribution Detection." *ICML*, 2023.

Elias Ramzi, Nicolas Audebert, Nicolas Thome, Clément Rambour, and Xavier Bitot. "Hierarchical Average Precision Training for Pertinent Image Retrieval." *ECCV*, 2022.

Elias Ramzi, Nicolas Thome, Clément Rambour, Nicolas Audebert, and Xavier Bitot. "Robust and Decomposable Average Precision for Image Retrieval." *NeurIPS*, 2021.